

ARCHIP

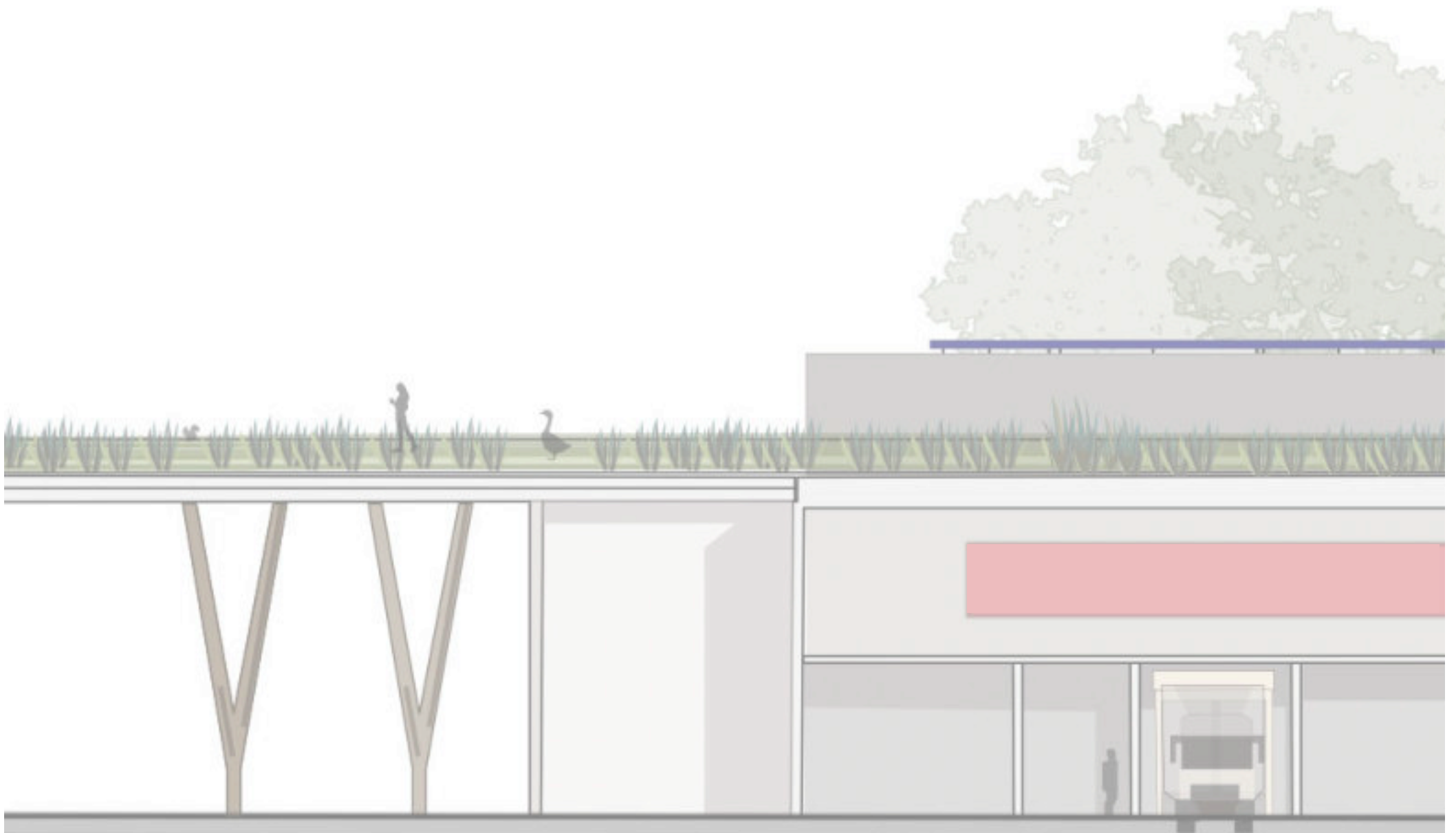
RETHINKING LOGISTICS

Kruilla-Prokop studio

AD2

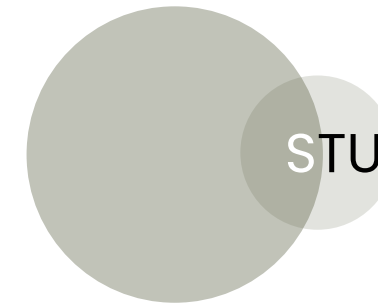
Ece YASAR

2026



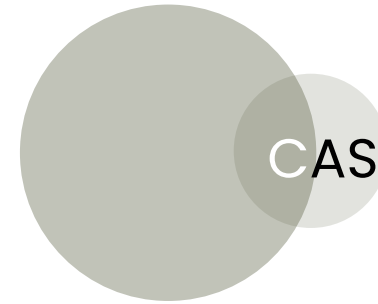


01



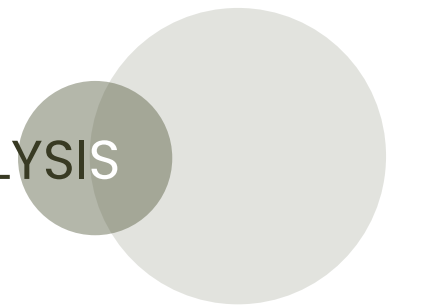
STUDIO

03



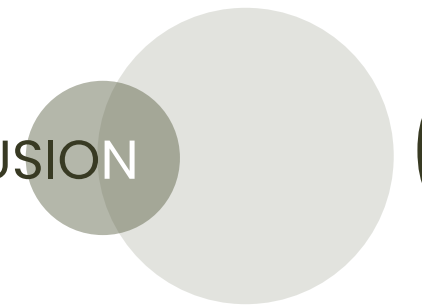
CASE STUDY

SITE ANALYSIS



02

CONCLUSION



04

O1

STUDIO

RETHINKING .LOGISTICS

This studio challenges the perception of logistics architecture as mere "big box" infrastructure. We view these sites not as isolated technical objects, but as active participants in our urban and ecological systems. In partnership with industry leaders like Ramboll, Henning Larsen, and Panattoni, students will tackle real-world design hurdles.

By blending speculative design with evidence-based research, we will explore how logistics hubs can transition from mono-functional shells into multi-layered, adaptable systems that generate measurable value for biodiversity, energy efficiency, and the local community.

RESEARCH

Green Roof as Landscape Tool — Macallan Distillery, Speyside UK, 2018.

A large-scale extensive green roof using native meadow grasses and wildflower planting to visually dissolve an industrial building into its surrounding terrain.

Demonstrates how substrate depth, plant selection, and roof geometry can make infrastructure disappear into landscape rather than sit on top of it.



Terrain Integration and Slope Construction — Preci Factory, Italy, 2015. Building mass set into the hillside using rammed earth and local stone, with a green roof continuing the ground plane over the structure. Rainwater harvested through a retention basin integrated as an architectural element. Shows material honesty and site-specific construction as ecological strategy.



Biosolar Roof System — Combined green roof and photovoltaic installation using low-growing shade-tolerant sedum and wildflower substrate between panel rows. Vegetation cools panels below the 25°C efficiency threshold while panels create microhabitat through shade and shelter. A dual-performance material system applicable directly to the Walther roof.

LCA

What is LCA?

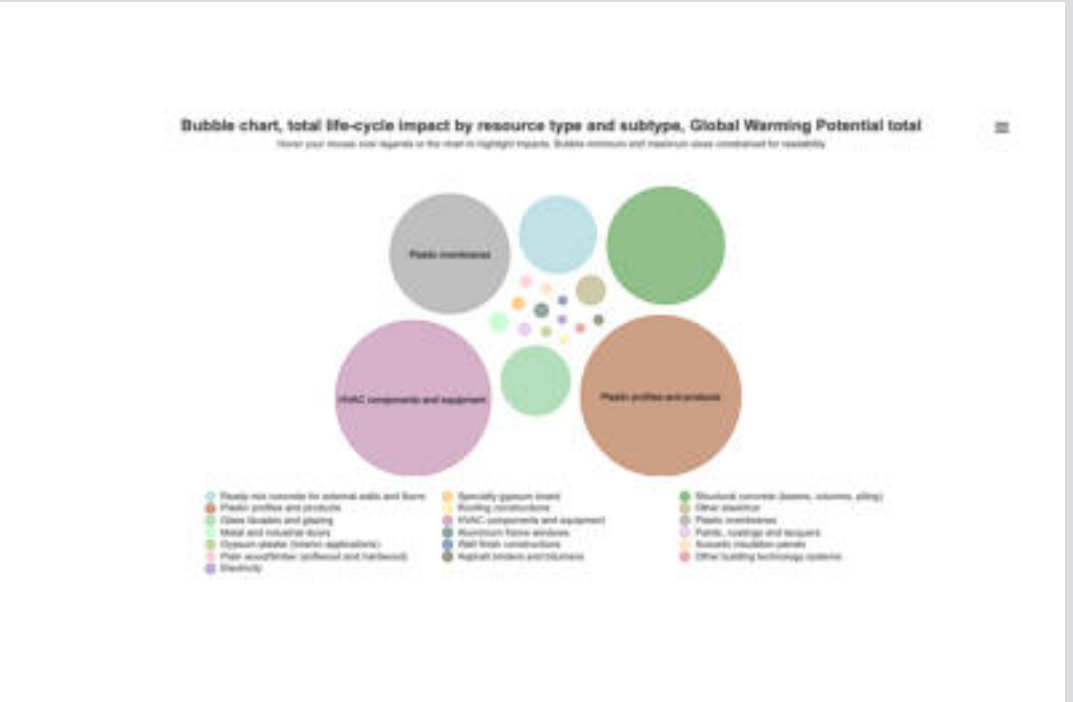
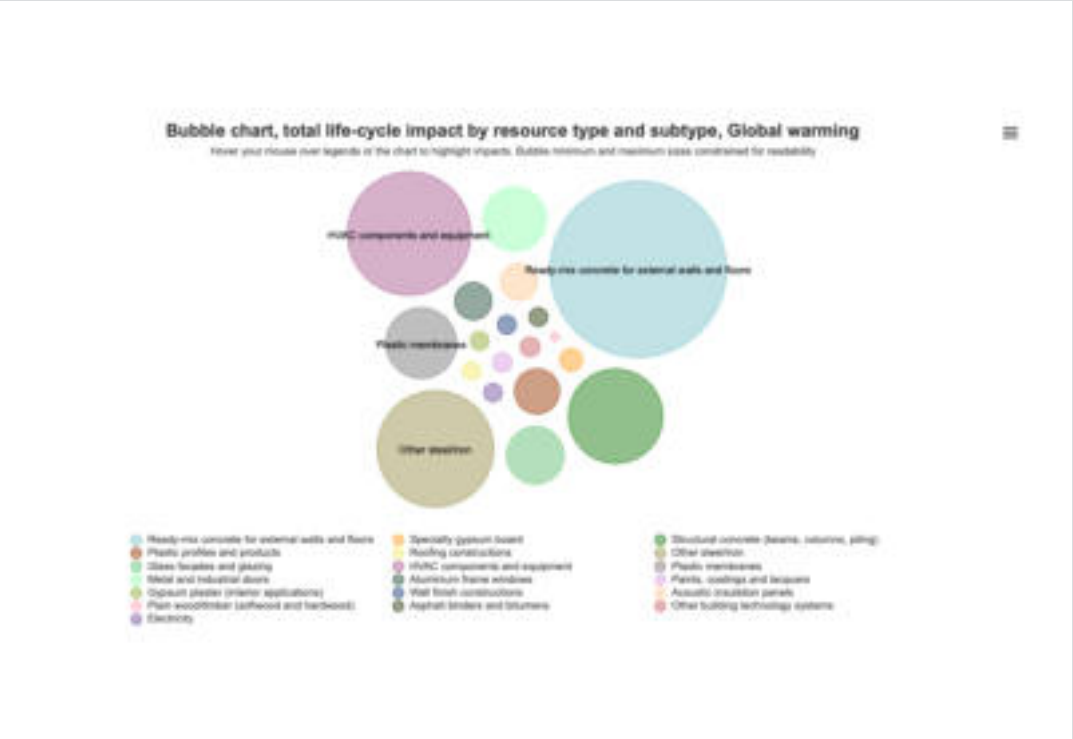
Life Cycle Assessment measures the environmental impact of a building from raw material extraction through construction, use and demolition. It makes performance visible, quantifying carbon, biodiversity stress, energy and water across the building's full lifespan.

How we used it?

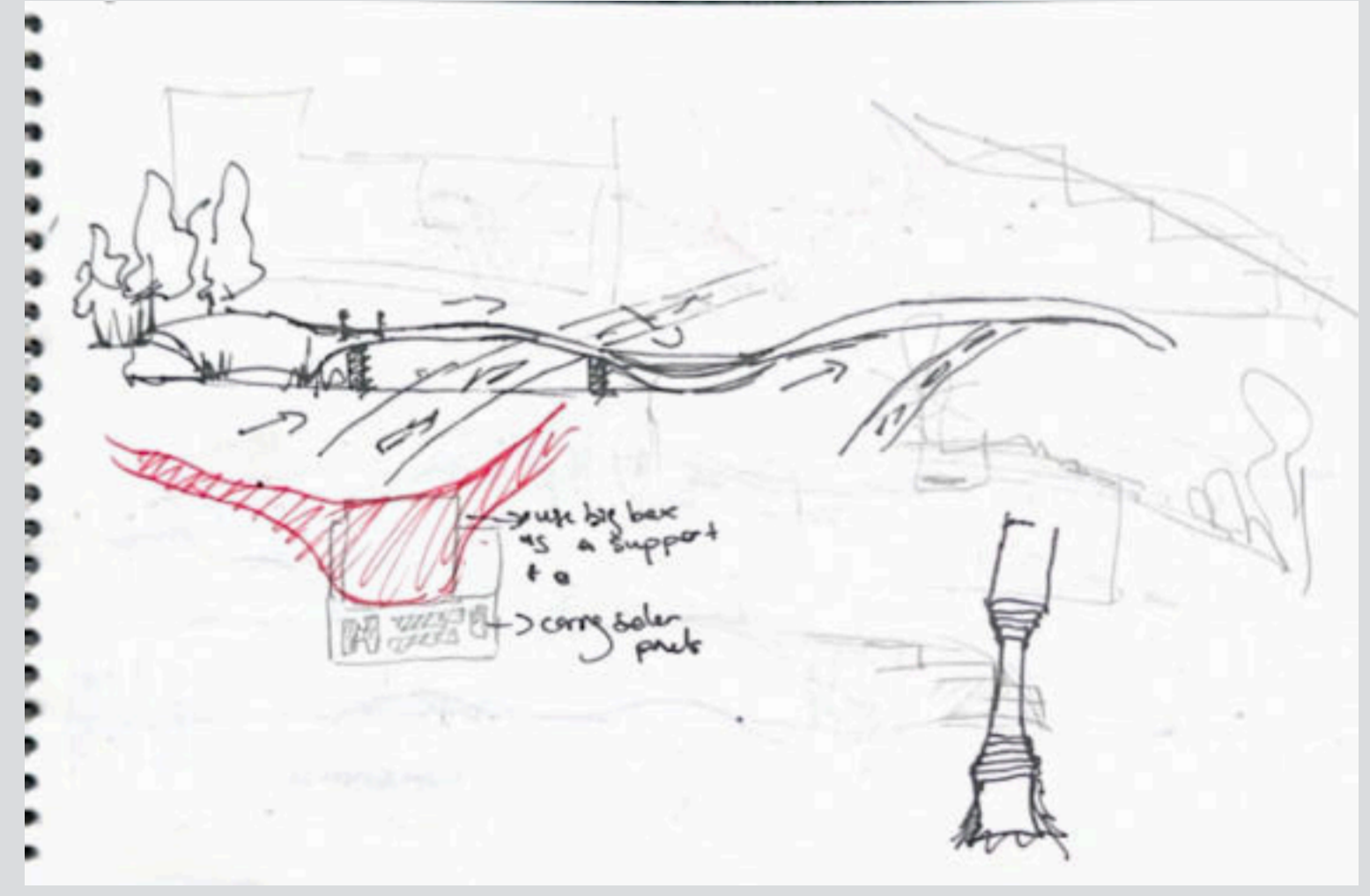
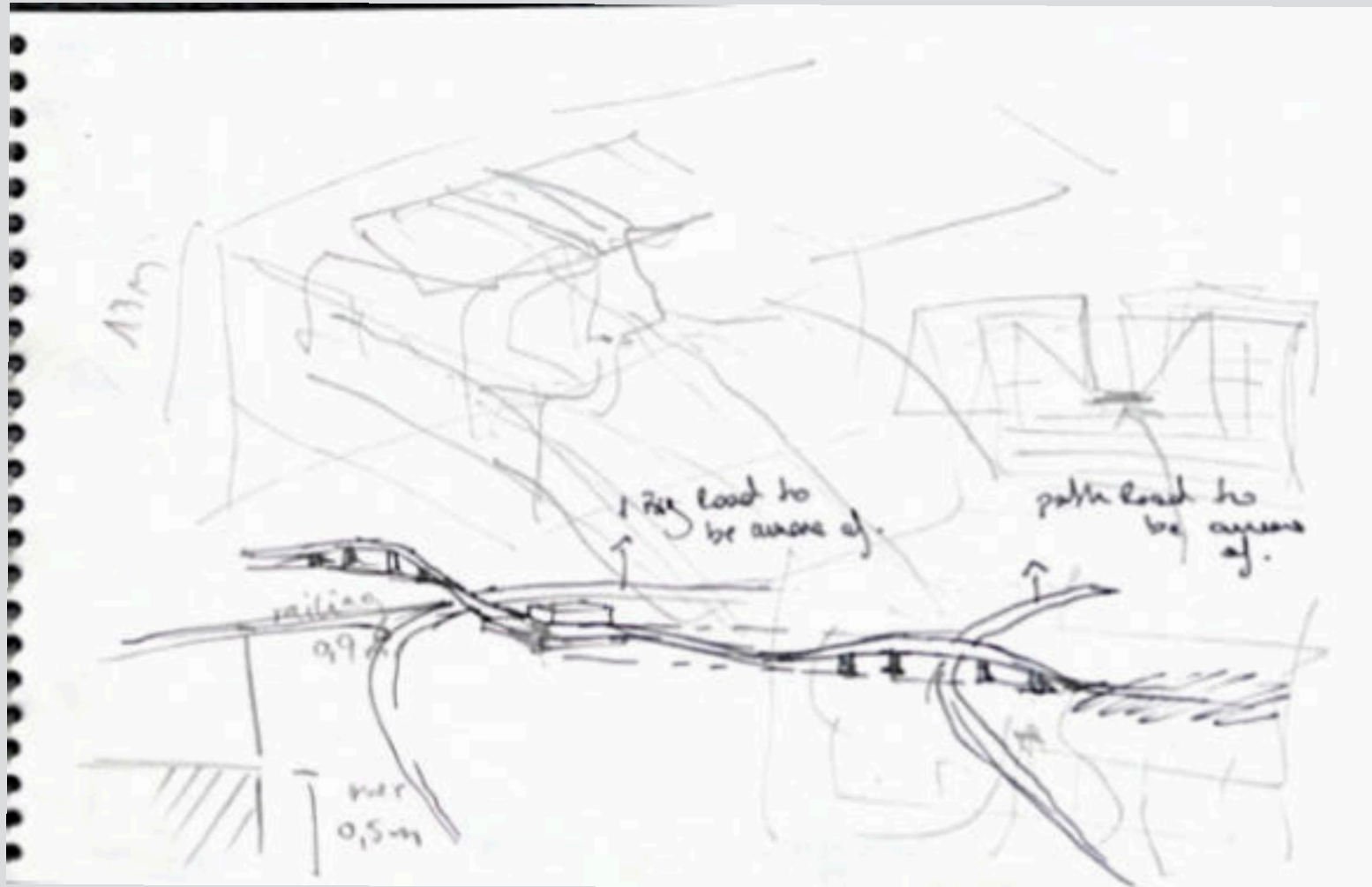
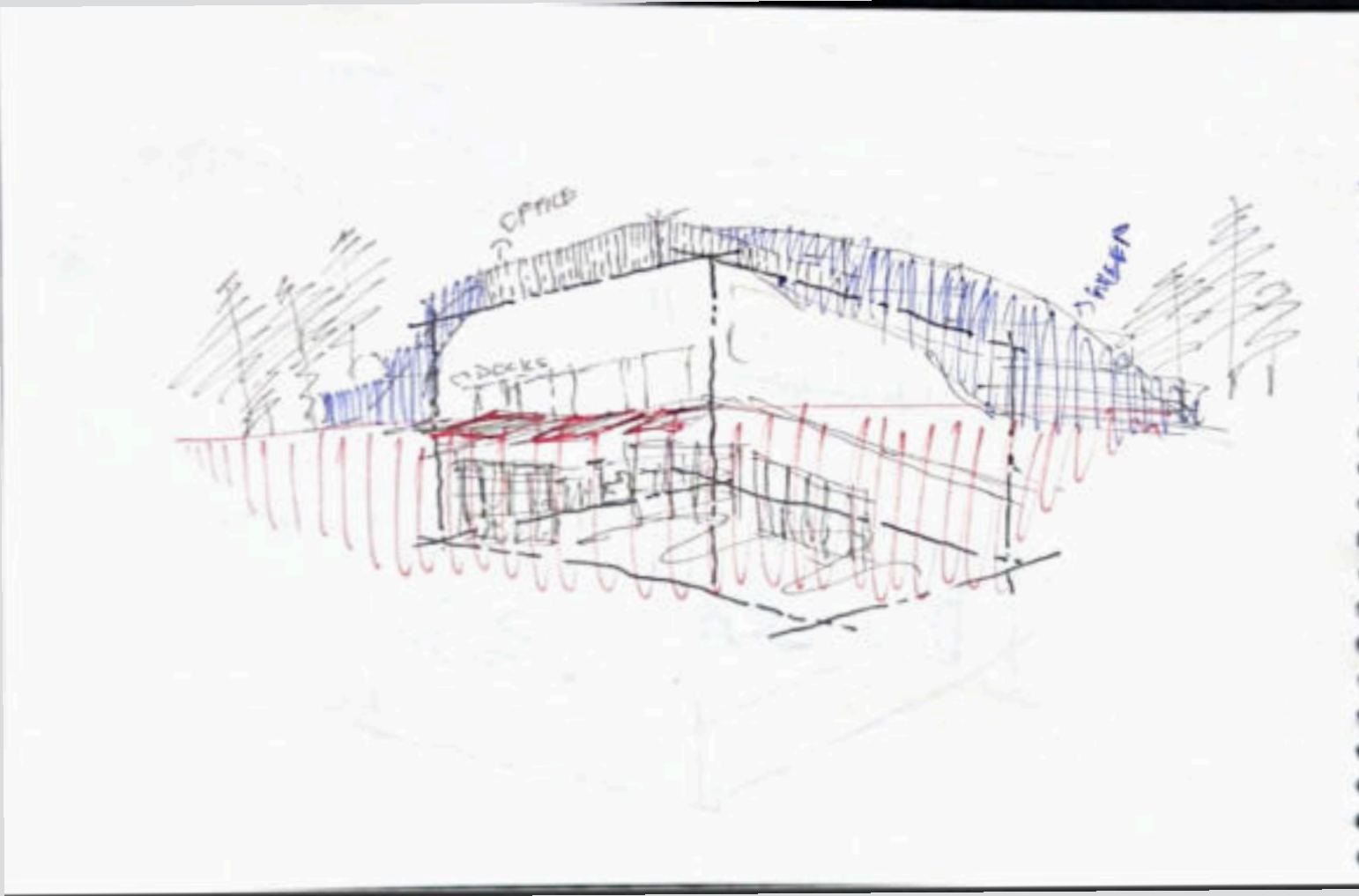
One Click LCA was used to model the Walther cross-dock facility in Kitzingen, assessed across construction stages A1–A5. Two tools ran in parallel : a standard LCA following EN15804, and a Biodiversity Stress tool measuring indirect ecological impacts through the supply chain.

What the results show?

The bubble charts make the problem clear. Concrete, HVAC systems and plastic membranes are responsible for the majority of the building's carbon footprint. In the full lifecycle assessment, plastics dominate materials carry a disproportionate long-term burden through replacement cycles. The clearest argument for rethinking what a warehouse is made of.



SKETCHES





KITZINGEN

Kitzingen, Bavaria. A small logistics node on the edge of a forest,

Walther's cross-dock facility sits between the city and the trees,

receiving and dispatching goods along

the corridor to Würzburg.



STRENGTHS

- Strategic position on the Würzburg corridor compact logistics node with strong road connectivity.
- Two dense forest patches directly adjacent to the site , rare ecological asset within an industrial zone.
- Existing solar panels on the warehouse, energy infrastructure already in place.
- Small scale of the Walther facility, compact footprint with high potential for transformation.

WEAKNESSES

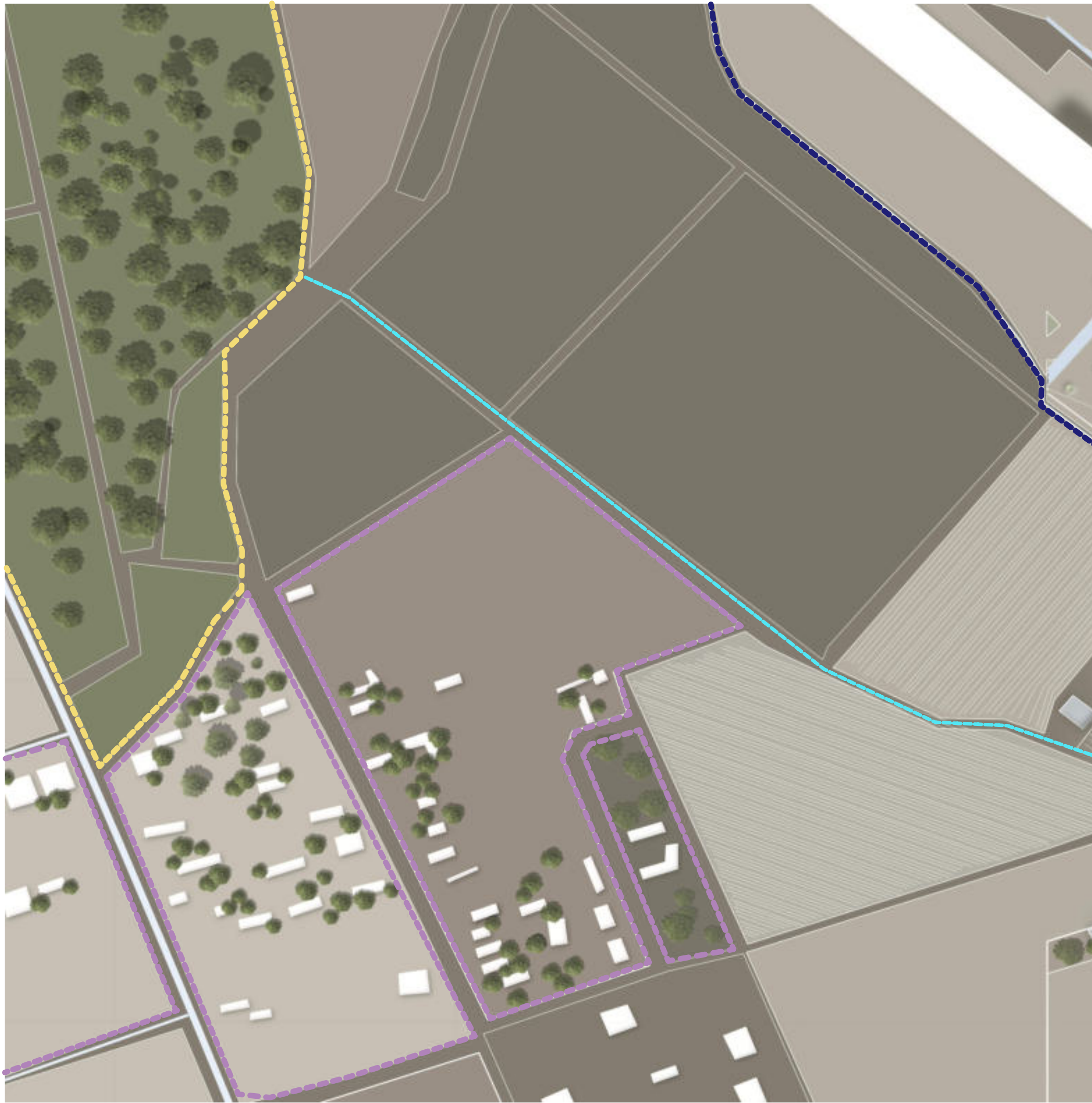
- Two major roads have severed the landscape, wildlife movement completely blocked at ground level.
- Warehouse roof entirely unused, dead concrete surface contributing nothing ecologically or socially.
- Hard industrial boundary with no green buffer, abrupt edge between nature and infrastructure.
- No provision for employees, no shade, no green space, no sustainable commute infrastructure.

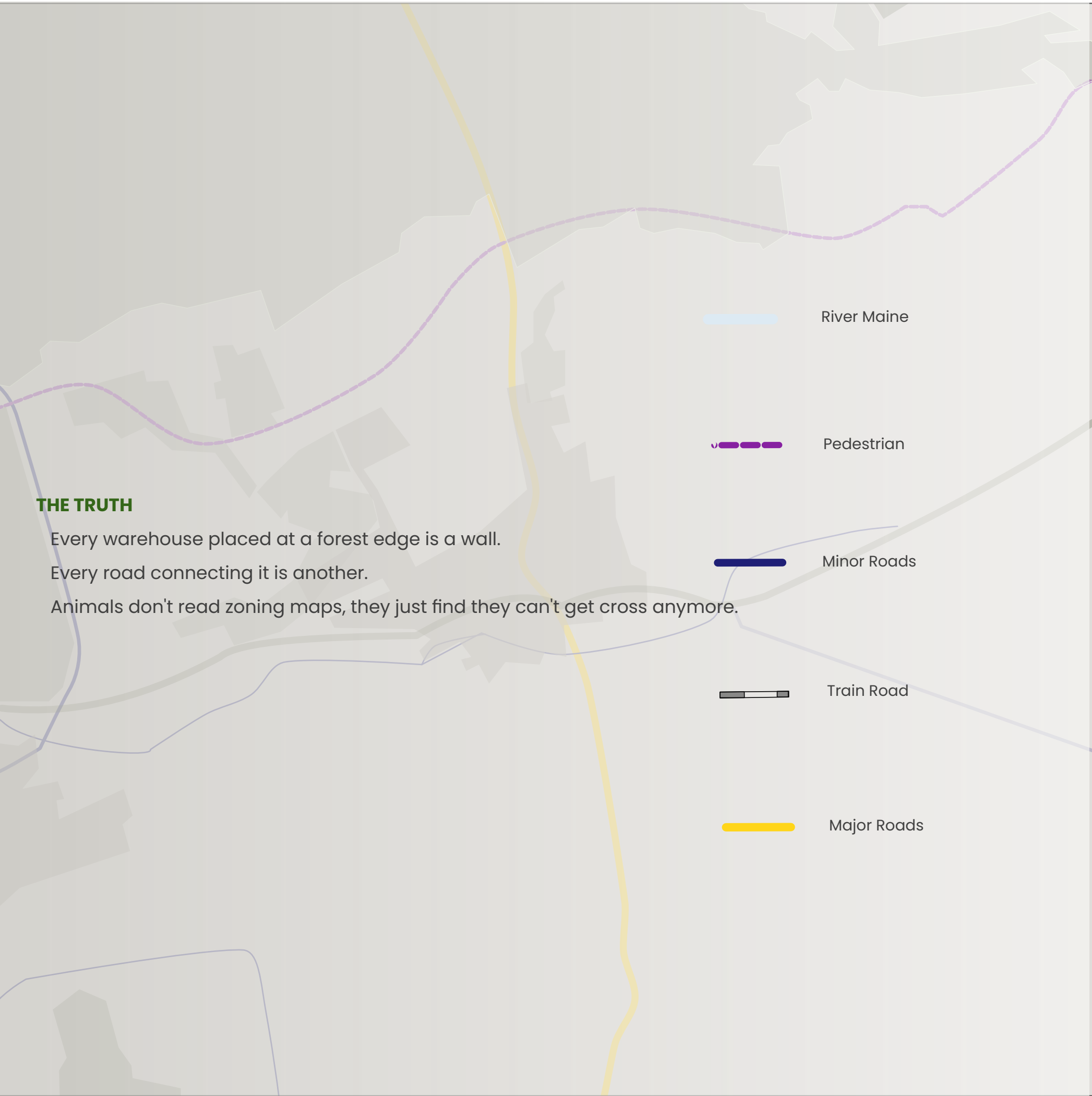
OPPORTUNITIES

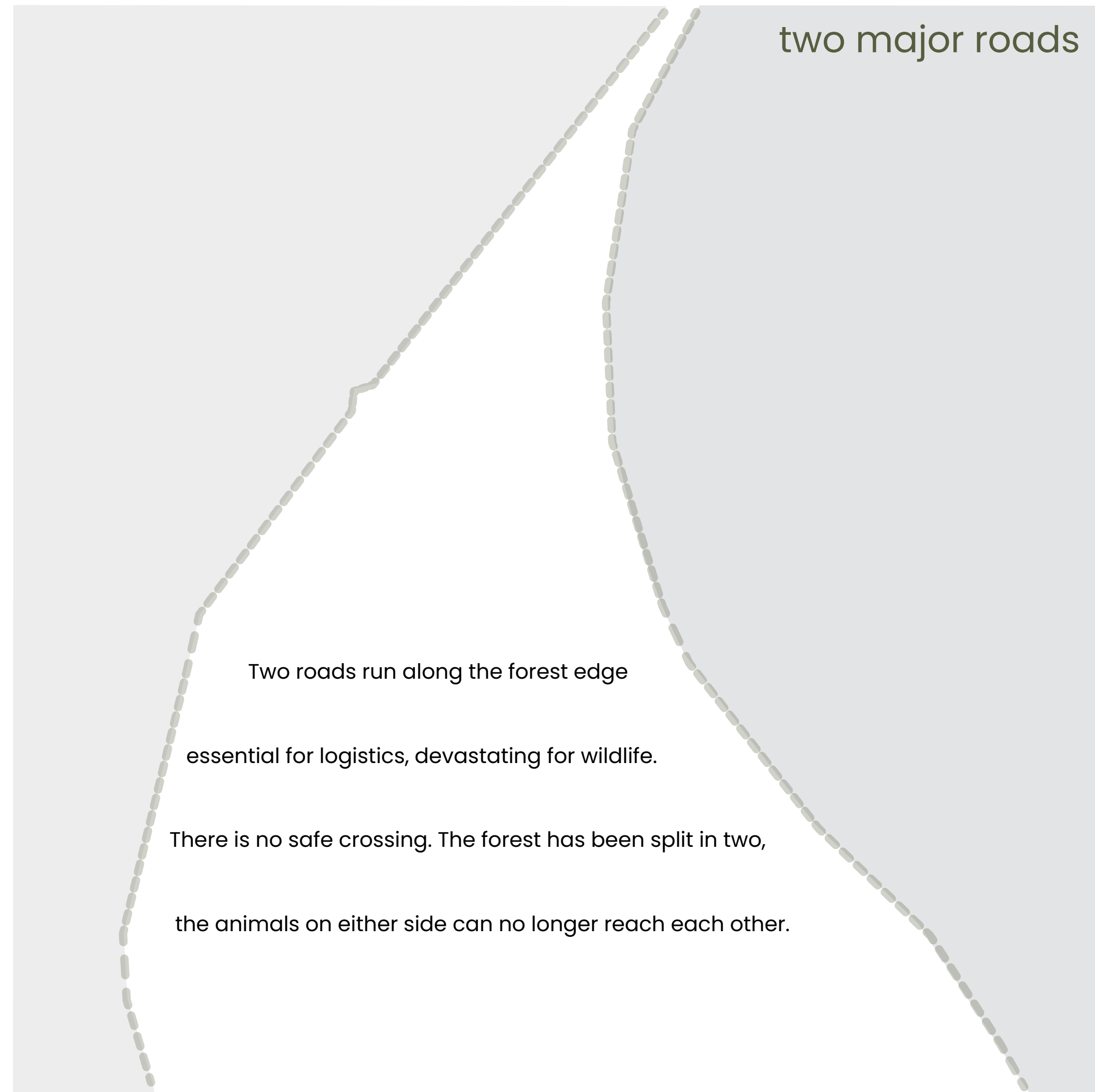
- Two forest patches close enough to reconnect, the building sits at exactly the right point to act as a bridge.
- Rooftop scale large enough to function as a meaningful ecological platform.
- Growing pressure on logistics industry to meet ESG and BREEAM standards.
- Residential neighbourhood nearby, potential to create shared green value beyond the site boundary.

NEEDS

- A wildlife crossing that goes over rather than through the road barrier.
- Native green roof to restore habitat lost to industrial development.
- Human-scale amenity; shade, cycling infrastructure, green edges for workers.
- A softer boundary between the building and the forests on either side.







03

CASE STUDY

WHAT IS BUILT?

Logistics infrastructure is everywhere. It keeps our cities running, our shelves stocked, our lives moving. But somewhere along the way, we drew a hard line between industry and nature and nature lost. Roads cut through forests. Warehouses landed on green edges. Wildlife corridors disappeared without anyone really noticing.



Every logistics facility near green edges
is an opportunity waiting to be rethought.



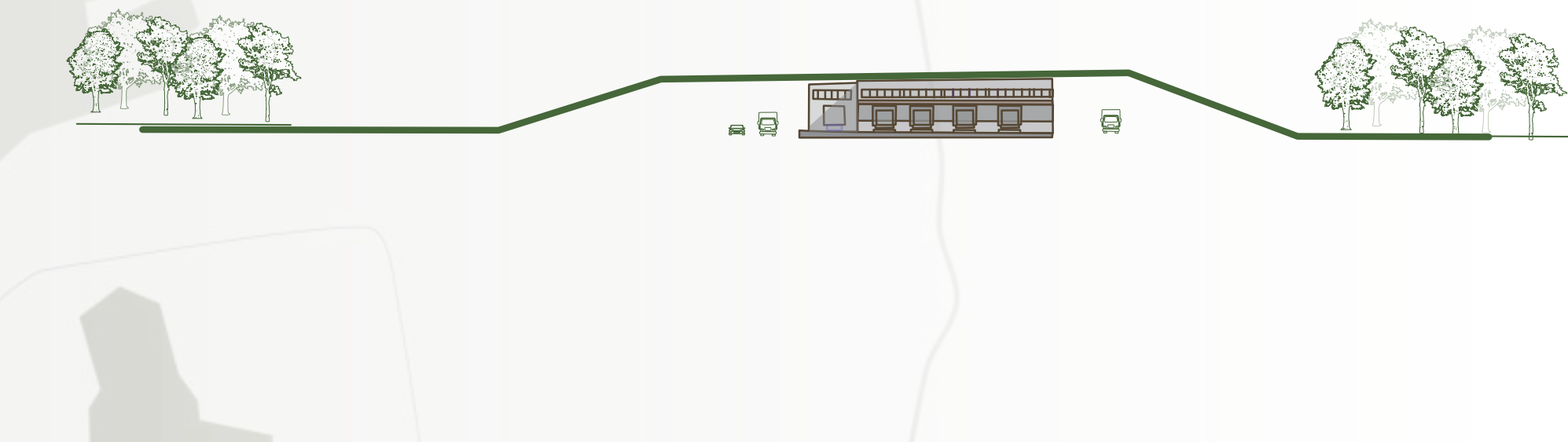
By treating the rooftop as ecological connection, a logistics building can stitch
fragmented forest patches back together.

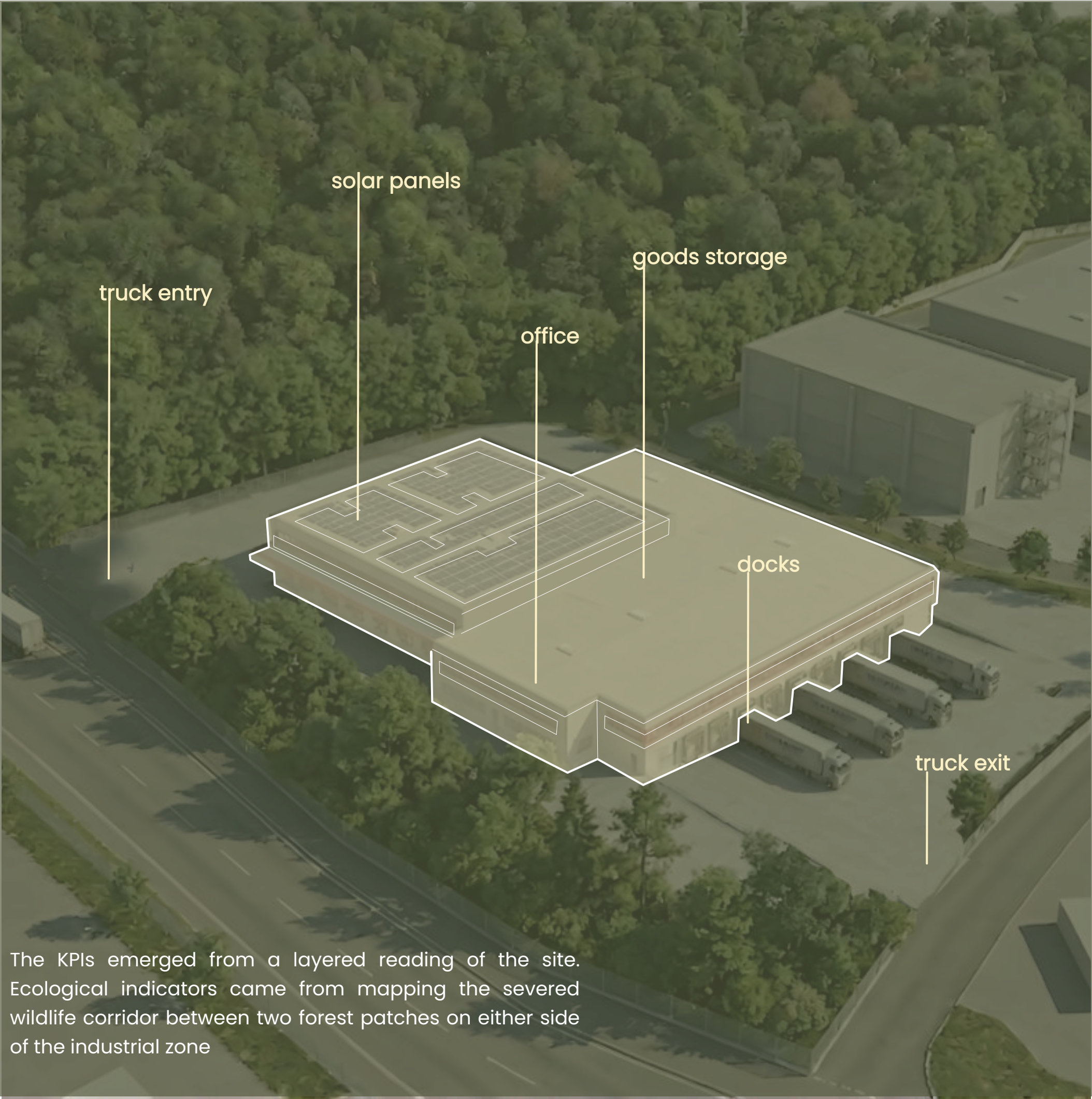
Native planting.

Elevated corridors.

Wildlife moving freely above the roads that once stopped them.

The building doesn't disappear ,it just finally starts pulling its weight for the
ecosystem around it.

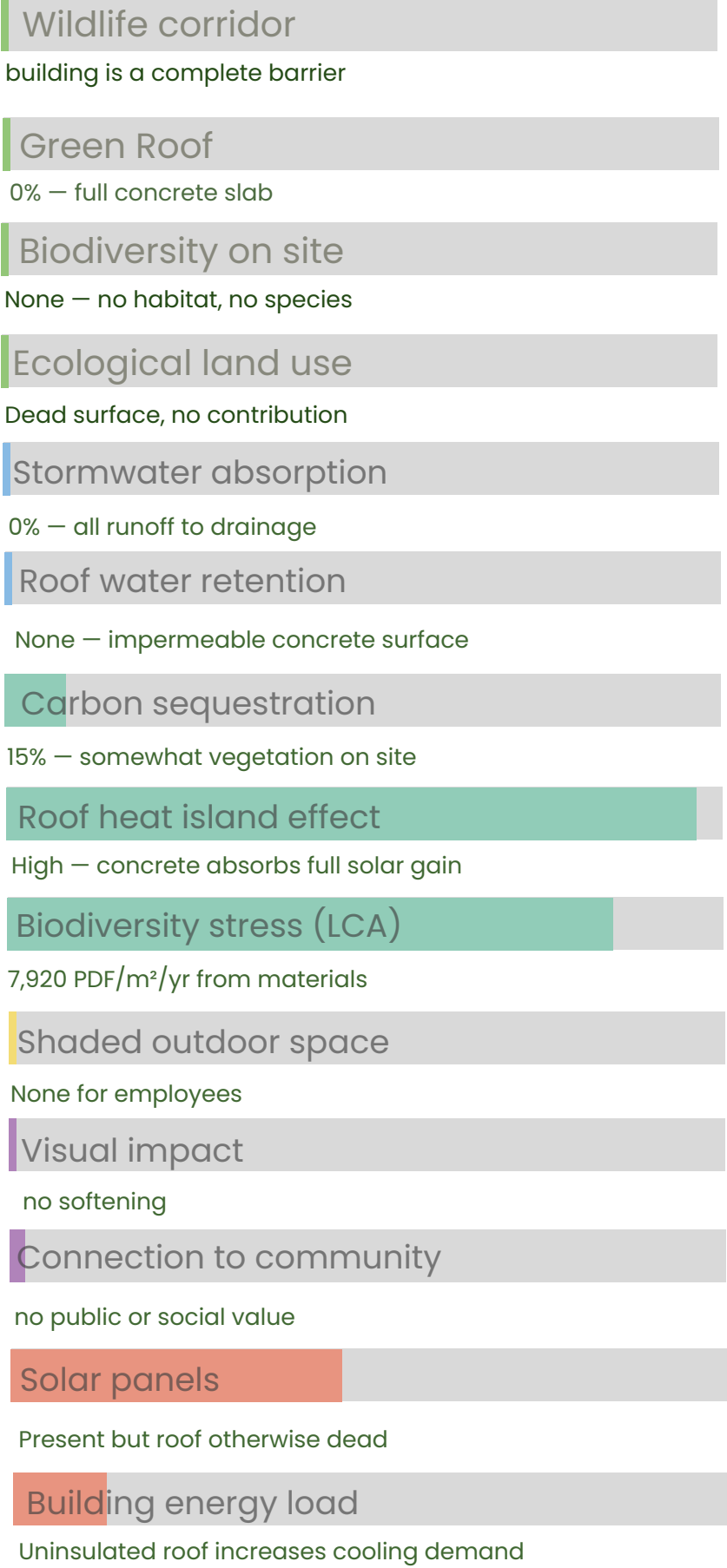




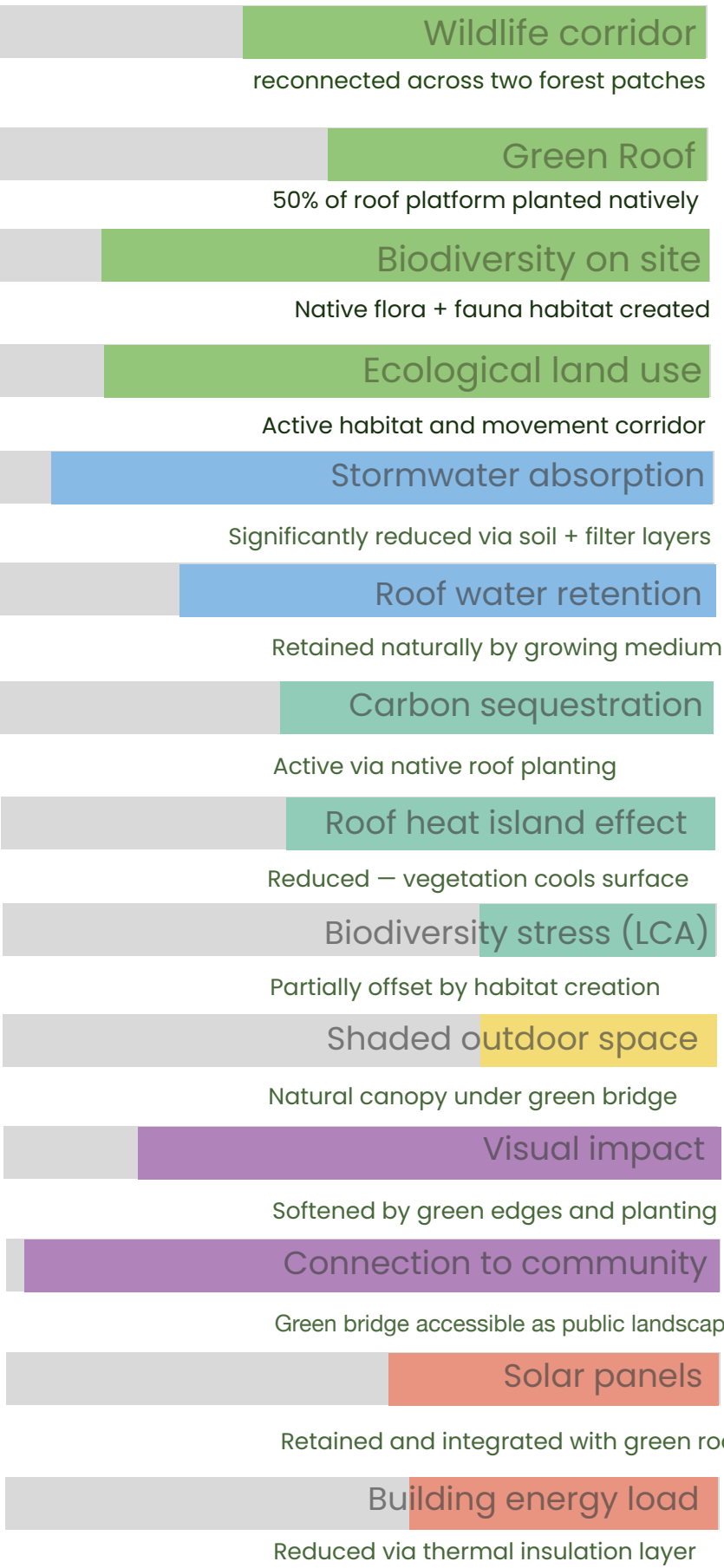
The KPIs emerged from a layered reading of the site. Ecological indicators came from mapping the severed wildlife corridor between two forest patches on either side of the industrial zone

KPIs

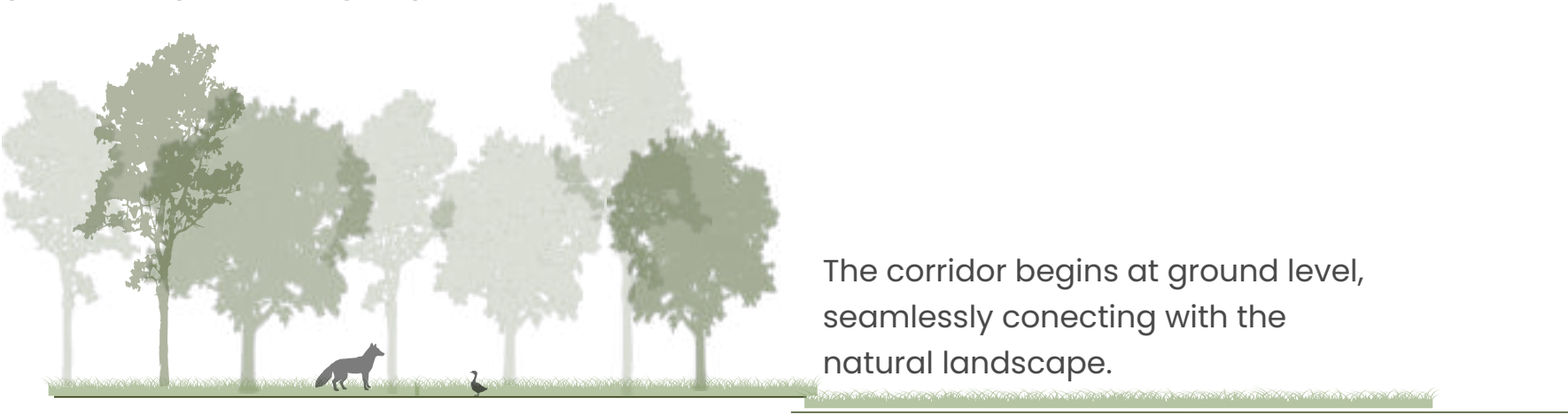
PROBLEMS



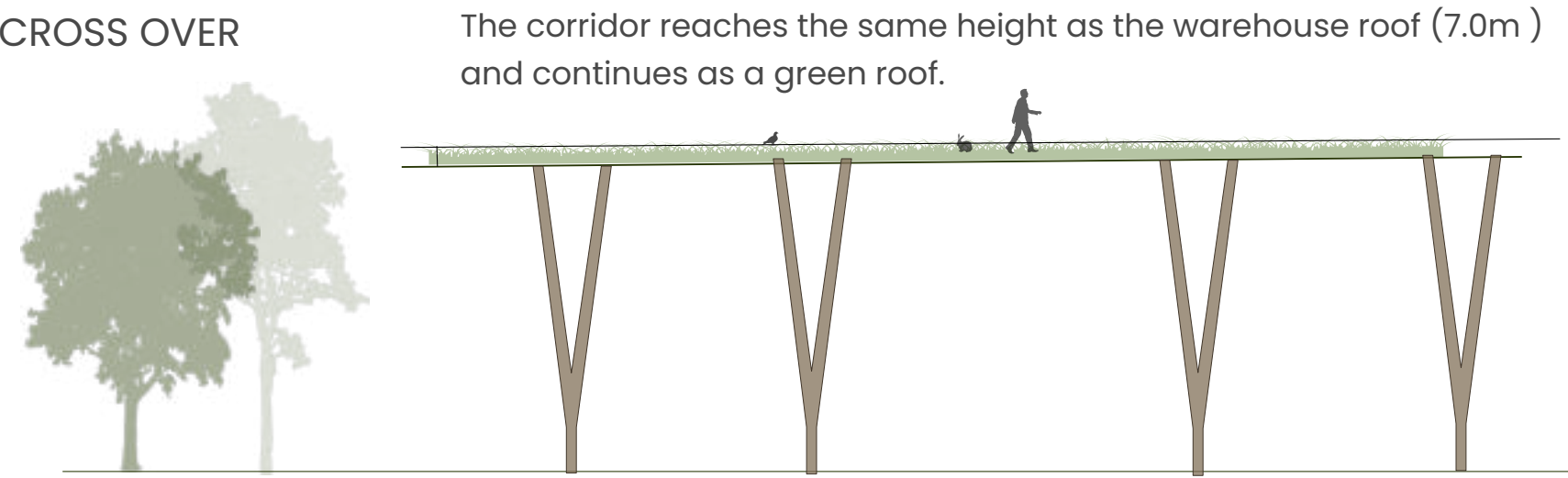
SOLUTIONS



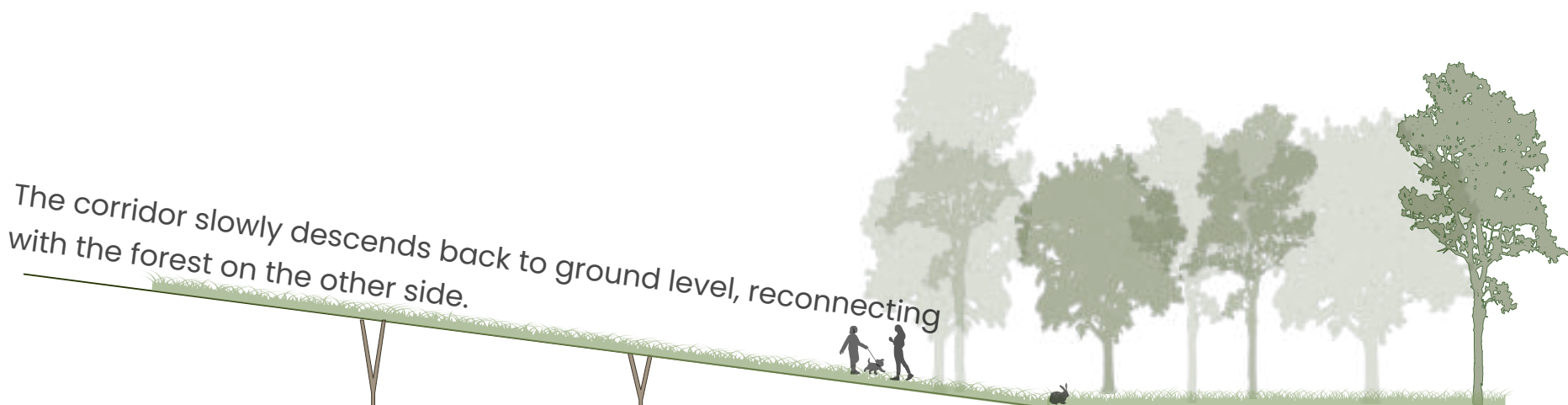
START FROM THE FOREST



CROSS OVER

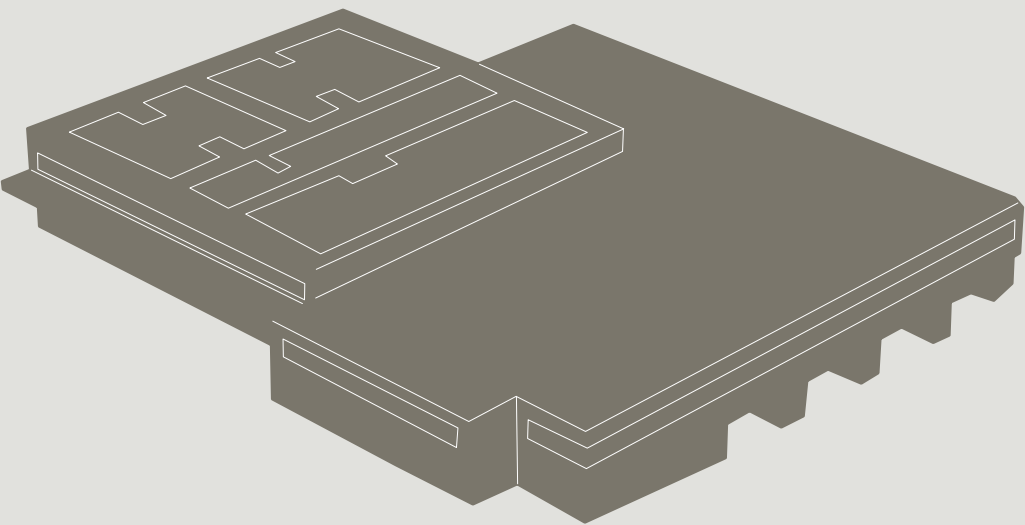


GRADUAL RISE



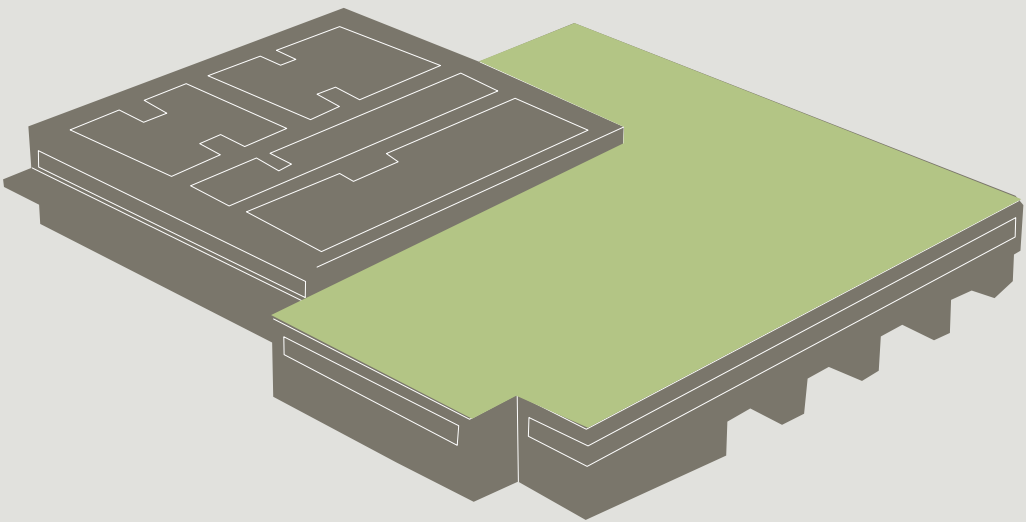
DESCEND TO THE OTHER FOREST

DEAD SURFACE



The roof is just a roof. A flat, concrete slab absorbing heat all day and dumping rainwater straight into the drainage system. It contributes nothing to its surroundings no habitat, no cooling, no life. From above, it reads as dead space. A missed opportunity sitting right at the treetop level of the forest next to it.

LIVING SURFACE

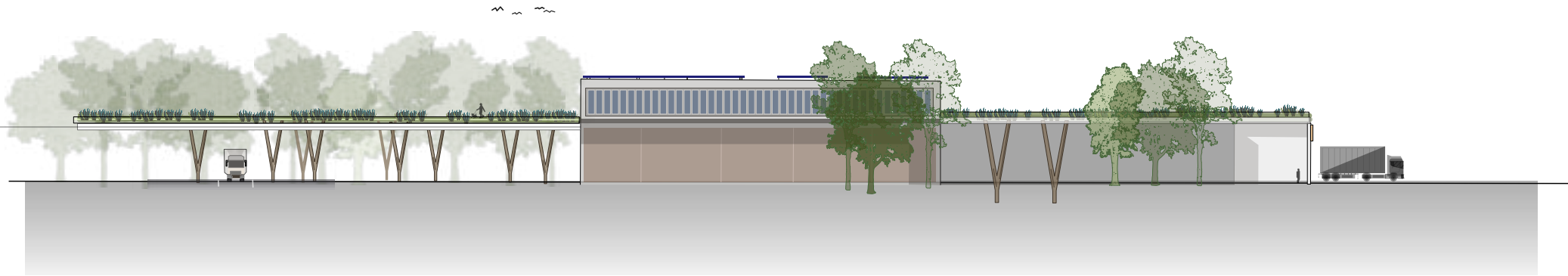


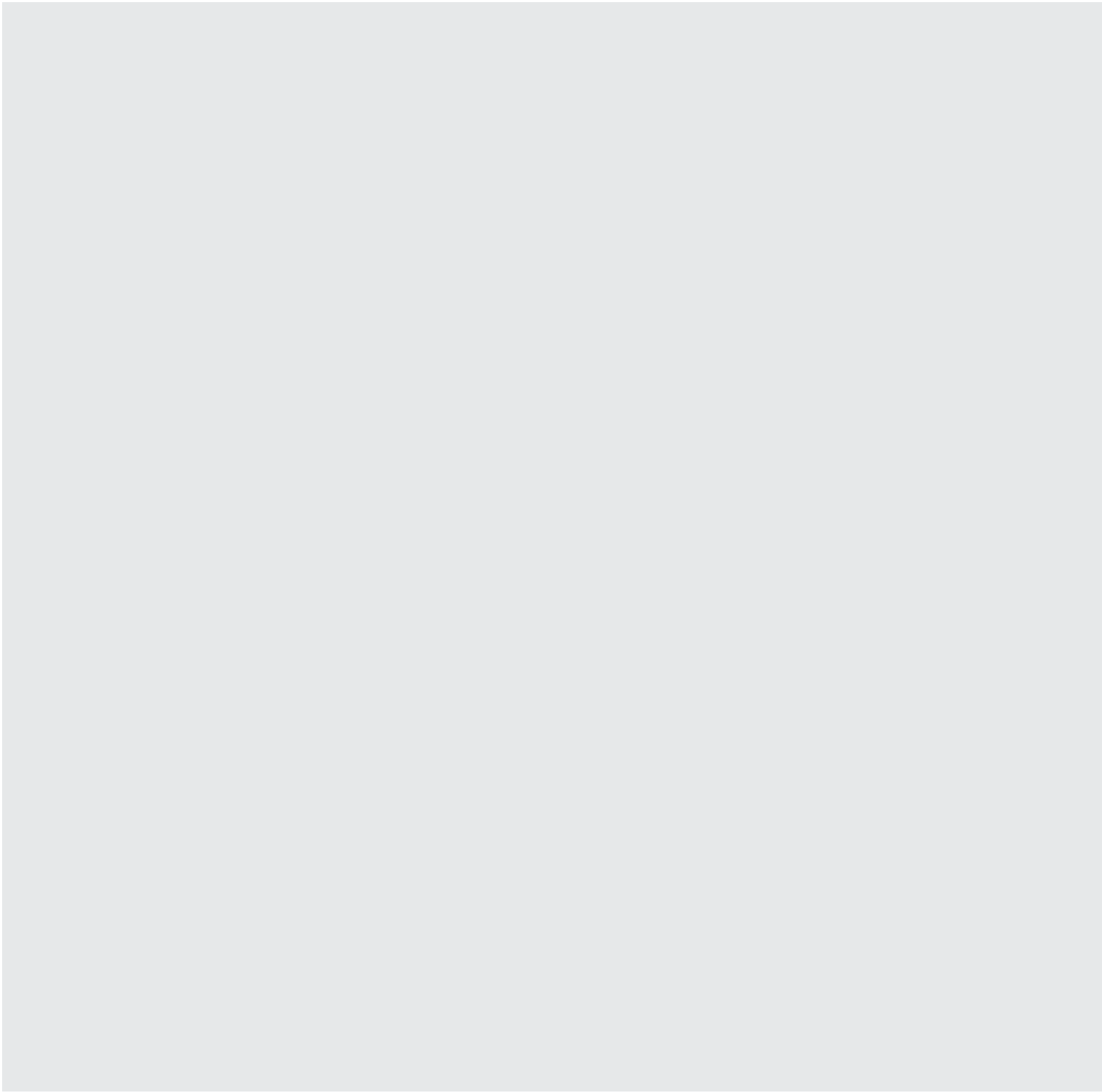
The same roof becomes the most active part of the building. Native plants replace concrete. Rainwater gets absorbed naturally, reducing runoff significantly. The surface cools down, taking pressure off the building's energy consumption. But more importantly it becomes habitat. Insects, birds and small mammals find a new patch of green that didn't exist before. The roof stops being a barrier between the building and the forest and starts being an extension of it. From above, you can barely tell where the forest ends and the building begins.

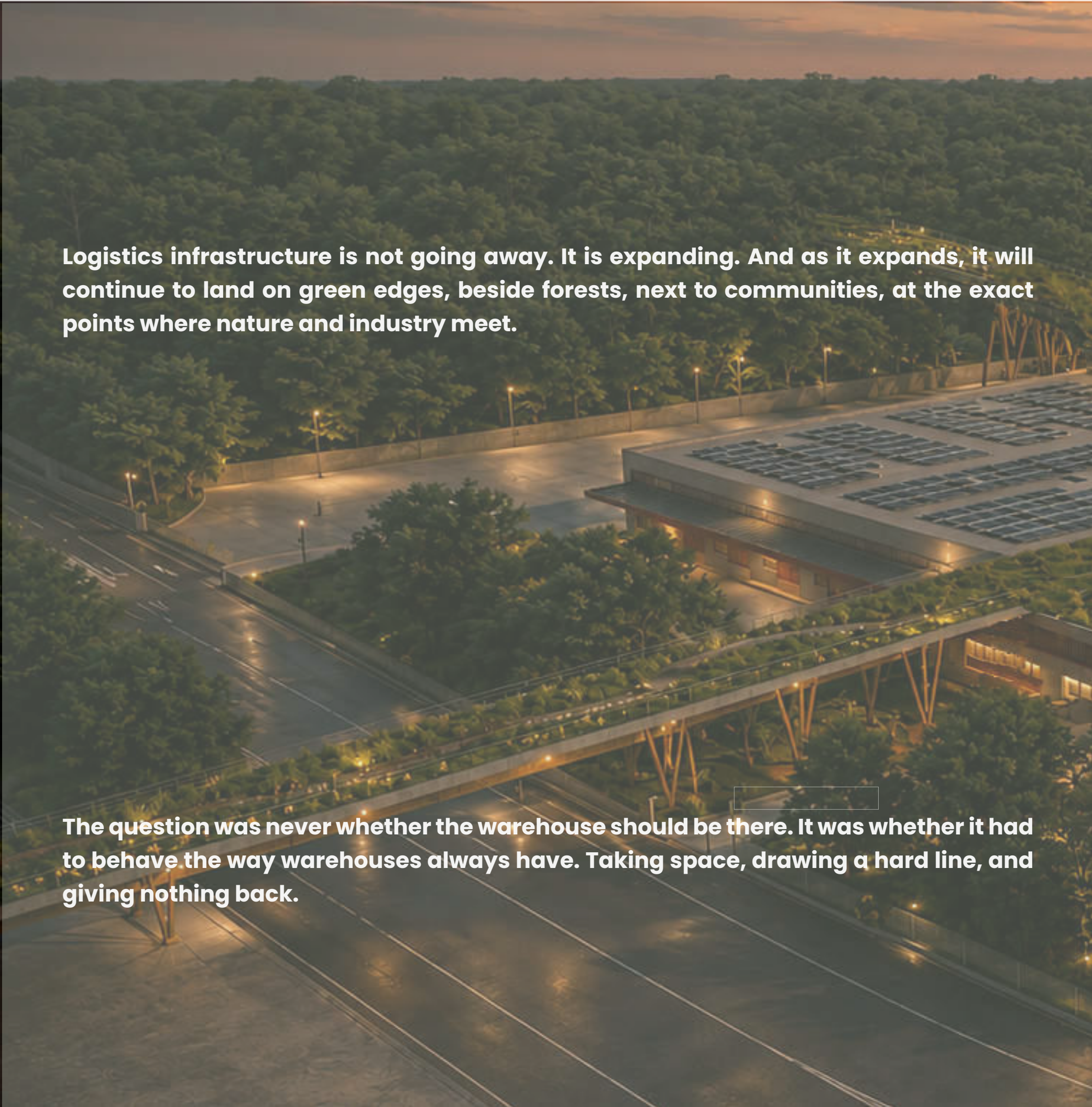
NORTH ELEVATION



WEST ELEVATION





An aerial photograph of a modern industrial building at dusk. The building features a large, flat roof covered in greenery. A bridge with a wooden support structure spans a road in front of the building. The surrounding area is lush with trees, and the sky is a mix of orange and blue.

Logistics infrastructure is not going away. It is expanding. And as it expands, it will continue to land on green edges, beside forests, next to communities, at the exact points where nature and industry meet.

The question was never whether the warehouse should be there. It was whether it had to behave the way warehouses always have. Taking space, drawing a hard line, and giving nothing back.

An aerial photograph of a large industrial building at dusk. The building has a green roof and a red sign that reads "WALTHER". Several semi-trucks are parked in front of the building. The surrounding area is lush with trees, and the sky is a mix of orange and blue.

Walther's cross-dock in Kitzingen became the test case. A compact building in a charged location. Two forests on either side. Two roads cutting through. A residential neighbourhood watching from below.

The proposal did not demolish or replace. It rethought. The roof became a platform. The platform became a bridge. The bridge reconnected what the roads had severed. Wildlife got a corridor. The building stopped being a barrier and started being part of the landscape it sat in.

ARCHIP

RETHINKING LOGISTICS

Kruilla-Prokop studio

AD2

Ece YASAR

2026